The Extension Graph of Spoof Lehmer Factorizations Is a Forest

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Abstract

Hasanalizade recently showed that, given a spoof factorization F satisfying $k\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$ for $\alpha = \pm 1$, the search for two-factor extensions $F \cdot x \cdot y$ satisfying the k-Lehmer condition $k\tilde{\varphi}^*(F \cdot x \cdot y) = \tilde{\epsilon}(F \cdot x \cdot y) - 1$ reduces to a single integer factorization. We study the global structure that this extension operation imposes on the set of k-Lehmer factorizations. We prove three results: (i) positive-integer extensions arise exclusively from $\alpha = +1$ seeds, (ii) the descent formula relating the evaluation of a Lehmer factorization to any descended pair is *independent of* k, and (iii) the resulting extension graph $\mathcal{G}_k$ is a **forest** for *every* $k \geq 2$—that is, no k-Lehmer factorization with factors ≥ 2 admits more than one descended pair. A computational census for $k = 2$ classifies all 39 factorizations reachable through seeds with at most six factors, identifies exactly 9 primitive factorizations, and reveals 17 plus-seeds, of which only six belong to the Fermat partial-product chain.

1 Introduction

Molnar and Singh [4] introduced the study of *spoof Lehmer factorizations*: multisets $F = \{x_1, \ldots, x_r\}$ of positive integers satisfying

$$k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) - 1, \qquad k \in \mathbb{Z}_{>0}, \tag{1}$$

where $\tilde{\epsilon}(F) = \prod x_i$ and $\tilde{\varphi}^*(F) = \prod(x_i - 1)$. Hasanalizade [1] proved that if a factorization F satisfies the auxiliary condition $k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$ for $\alpha \in \{-1, +1\}$, then the search for two-factor extensions $F \cdot x \cdot y$ satisfying (1) reduces to factoring a single explicit integer (Equation (2) below).

This note investigates the global structure that the Hasanalizade extension imposes on the solution set. The main contributions are:

(i) A proof that only $\alpha = +1$ seeds can produce extensions with positive factors (Theorem 2).

(ii) A closed-form descent criterion (Theorem 4) and the observation that it is *independent of* k (Corollary 5).

(iii) A proof that the extension graph $\mathcal{G}_k$ is a forest for every $k \geq 2$ (Theorem 8).

(iv) A computational census for $k = 2$ classifying primitives and plus-seeds (Theorem 13).

2 Preliminaries

We follow the notation of [4] and [1]. A *spoof factorization* is a finite multiset $F = x_1 \cdots x_r$ of positive integers with $x_i \geq 2$, with *spoof evaluation* $\tilde{\epsilon}(F) = \prod x_i$ and *spoof unitary totient* $\tilde{\varphi}^*(F) = \prod(x_i - 1)$. We write $|F| = r$ for the number of factors. A factorization F with $|F| \geq 2$ is *unitary k-Lehmer* if (1) holds.

Definition 1. A factorization F is an *α-seed* (with respect to k) if $k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$, where $\alpha \in \{-1, +1\}$. When $\alpha = +1$, we call F a *plus-seed*.

Hasanalizade's Theorem 3 in [1] states that if F is an α-seed with $N = \tilde{\epsilon}(F)$, then $F \cdot x \cdot y$ is k-Lehmer if and only if

$$(\alpha x - N - \alpha)(\alpha y - N - \alpha) = N^2 + \alpha N - \alpha. \tag{2}$$

We also use the *deficiency function* $g(F) = k\,\tilde{\varphi}^*(F) - \tilde{\epsilon}(F)$, which satisfies the recurrence

$$g(F \cdot a) = (a-1)\,g(F) - \tilde{\epsilon}(F), \qquad g(\emptyset) = k - 1. \tag{3}$$

Lehmer factorizations correspond to $g = -1$, plus-seeds to $g = 1$.

3 Only plus-seeds produce positive extensions

Theorem 2. *Let $k \geq 2$ and let F be a (-1)-seed with $\tilde{\epsilon}(F) = N \geq 1$. Then Equation (2) has no solutions with $x, y \geq 2$.*

Proof. For $\alpha = -1$, Equation (2) becomes $(x + N - 1)(y + N - 1) = N^2 - N + 1$. If $x, y \geq 2$ then the left side is at least $(N+1)^2 = N^2 + 2N + 1 > N^2 - N + 1$ for all $N \geq 1$. □

4 The descent formula is k-independent

Definition 3. Let G be a k-Lehmer factorization. An unordered pair $\{a, b\}$ of factors of G (counted with multiplicity) is a *descended pair* if $F = G \setminus \{a, b\}$ is a plus-seed from which G arises via Equation (2). We call G *primitive* if it admits no descended pair, and *derived* otherwise.

Theorem 4 (Descent formula)**.** *Let $k \geq 2$ and let G be a k-Lehmer factorization with $\tilde{\epsilon}(G) = N$. An unordered pair $\{a, b\}$ of factors of G is a descended pair if and only if*

$$N(a + b - 1) = ab\big((a-1)(b-1) + 1\big). \tag{4}$$

Proof. The sub-factorization $F = G \setminus \{a, b\}$ has $\tilde{\epsilon}(F) = N/(ab)$ and $\tilde{\varphi}^*(F) = \tilde{\varphi}^*(G)/\big((a-1)(b-1)\big)$. Since G is k-Lehmer and F is a plus-seed:

$$k\,\tilde{\varphi}^*(F) = \frac{k\,\tilde{\varphi}^*(G)}{(a-1)(b-1)} = \frac{N-1}{(a-1)(b-1)}, \qquad k\,\tilde{\varphi}^*(F) = \frac{N}{ab} + 1.$$

Equating:

$$\frac{N-1}{(a-1)(b-1)} = \frac{N}{ab} + 1. \tag{5}$$

Clearing denominators: $(N-1)\,ab = N(a-1)(b-1) + ab(a-1)(b-1)$. Expanding $N(a-1)(b-1) = N(ab - a - b + 1)$ and collecting:

$$N(a + b - 1) = ab\big(ab - a - b + 2\big) = ab\big((a-1)(b-1) + 1\big). \qquad \square$$

Corollary 5 (k-independence)**.** *The descent formula* (4) *is independent of k. Equivalently, whether a pair $\{a, b\}$ can be a descended pair is determined entirely by the multiset of factors and the evaluation N, not by the value of k.*

Proof. Equation (5) shows that k cancels in the passage from the k-Lehmer and plus-seed conditions to (4). □

Remark 6. The k-independence is perhaps surprising: the Lehmer condition $k\tilde{\varphi}^* = \tilde{\epsilon} - 1$ and the plus-seed condition $k\tilde{\varphi}^* = \tilde{\epsilon} + 1$ both involve k explicitly, but dividing the first by the second at the level of a descended pair absorbs k into the relationship between $\{a, b\}$ and the remaining factors.

5 The extension graph is a forest

Definition 7. The *extension graph* $\mathcal{G}_k$ is a directed graph whose vertices are the k-Lehmer factorizations with factors ≥ 2. There is an edge $G \to F$ whenever $F = G \setminus \{a, b\}$ for some descended pair $\{a, b\}$.

Theorem 8 (Forest theorem)**.** *For every $k \geq 2$, the extension graph $\mathcal{G}_k$ is a forest: every k-Lehmer factorization with factors ≥ 2 admits at most one descended pair.*

By Corollary 5, the descent formula (4) is k-independent, so the entire proof reduces to showing that no single multiset G with $\tilde{\epsilon}(G) = N$ can have two distinct unordered pairs satisfying (4) with an appropriate divisibility and positivity structure. The proof uses three lemmas, covering the three possible cases for the intersection of two descended pairs.

Lemma 9 (Disjoint case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit two descended pairs $\{a, b\}$ and $\{c, d\}$ with $\{a, b\} \cap \{c, d\} = \emptyset$.*

Proof. Write $G = \{a, b, c, d\} \cup E$ with $\tilde{\epsilon}(E) = Q$ (where $Q = 1$ if $E = \emptyset$), so $N = abcdQ$. Applying the descent formula (4) to each pair and dividing by ab and cd respectively:

$$cdQ\,(a + b - 1) = (a-1)(b-1) + 1, \tag{6}$$

$$abQ\,(c + d - 1) = (c-1)(d-1) + 1. \tag{7}$$

Since $(a-1)(b-1)+1 \leq ab$ for $a, b \geq 2$ (with strict inequality unless one of a, b equals 1), multiplying gives

$$a^2b^2c^2d^2\,Q^2\,(a+b-1)(c+d-1) \leq a^2b^2c^2d^2,$$

whence $Q^2(a + b - 1)(c + d - 1) \leq 1$. Since $a + b - 1 \geq 3$ and $c + d - 1 \geq 3$, we get $Q^2 \leq 1/9 < 1$, so $Q < 1$.

If $E \neq \emptyset$ then $Q \geq 2$, a contradiction. If $E = \emptyset$ then $Q = 1$ and (6)–(7) become $cd(a + b - 1) = (a - 1)(b - 1) + 1$ and $ab(c + d - 1) = (c - 1)(d - 1) + 1$. The first gives $cd < ab/(a + b - 1) \leq ab/3$; the second gives $ab < cd/(c + d - 1) \leq cd/3$. Together: $cd < ab/3 < cd/9$, which is absurd. □

Lemma 10 (Overlapping case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit two descended pairs $\{a, b\}$ and $\{b, d\}$ sharing one element, where $a \neq d$.*

Proof. Write $G = \{a, b, d\} \cup E$ with $\tilde{\epsilon}(E) = Q$, so $N = abdQ$. The descent formula for each pair gives

$$dQ\,(a + b - 1) = (a-1)(b-1) + 1, \tag{8}$$

$$aQ\,(b + d - 1) = (b-1)(d-1) + 1. \tag{9}$$

Bounding: $dQ \le ab/(a+b-1)$ and $aQ \le bd/(b+d-1)$. Multiplying:

$$adQ^2 \le \frac{ab^2 d}{(a+b-1)(b+d-1)}, \qquad \text{so} \qquad Q^2 \le \frac{b^2}{(a+b-1)(b+d-1)}.$$

We claim $(a+b-1)(b+d-1) > b^2$ for all $a, b, d \ge 2$. Expanding:

$$\begin{aligned}(a+b-1)(b+d-1) - b^2 &= a(b+d-1) + b(d-2) + (1-d) \\ &\ge 2(b+d-1) + 0 + (1-d) = 2b + d - 1 \ge 3.\end{aligned}$$

Therefore $Q^2 < 1$, which is impossible if $E \ne \emptyset$ (since then $Q \ge 2$).

If $E = \emptyset$, then $|G| = 3$, $Q = 1$, and $G = \{a, b, d\}$. The seed of $\{a, b\}$ is the single-element factorization (d), and the plus-seed condition $k(d-1) = d+1$ gives $d = (k+1)/(k-1)$. By symmetry, the seed of $\{b, d\}$ forces $a = (k+1)/(k-1)$. We consider three sub-cases:

- $k = 2$: $d = a = 3$, $G = \{3, b, 3\}$, $N = 9b$. The Lehmer condition $2 \cdot 2(b-1) \cdot 2 = 9b - 1$ gives $8b - 8 = 9b - 1$, hence $b = -7$.
- $k = 3$: $d = a = 2$, $G = \{2, b, 2\}$, $N = 4b$. The Lehmer condition $3 \cdot 1 \cdot (b-1) \cdot 1 = 4b - 1$ gives $3b - 3 = 4b - 1$, hence $b = -2$.
- $k \ge 4$: $(k+1)/(k-1)$ is not an integer (since $k - 1 \ge 3$ and $\gcd(k-1, k+1) \mid 2$), so no single-element plus-seed exists and the $|G| = 3$ case is vacuous.

In all cases, no valid factorization arises. □

Lemma 11 (Repeated-element case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit descended pairs $\{v, v\}$ and $\{v, w\}$ with $w \ne v$, where v appears with multiplicity at least two in G.*

Proof. Write $G = \{v, v, w\} \cup E$ with $\tilde{\epsilon}(E) = Q$, so $N = v^2 wQ$. The descent formula for the pair $\{v, v\}$ gives, after dividing by v^2:

$$wQ(2v-1) = (v-1)^2 + 1. \tag{10}$$

The left side is a positive multiple of $2v - 1$, so $(2v-1) \mid \big((v-1)^2 + 1\big)$. We compute modulo $2v - 1$: since $2(v-1) \equiv -1 \pmod{2v-1}$,

$$4\big((v-1)^2 + 1\big) = (2v-2)^2 + 4 \equiv 1 + 4 = 5 \pmod{2v-1}.$$

As $\gcd(4, 2v-1) = 1$, we conclude $(2v-1) \mid 5$, giving $2v - 1 \in \{1, 5\}$ and $v \in \{1, 3\}$.

For $v = 3$: Equation (10) gives $5wQ = 5$, so $wQ = 1$, forcing $w = 1$ and $Q = 1$. But factors must be ≥ 2.

For $v = 1$: factors must be ≥ 2.

Neither case yields a valid factorization. □

Proof of Theorem 8. Let G be a k-Lehmer factorization with factors ≥ 2, and suppose for contradiction that $\{a, b\}$ and $\{c, d\}$ are two distinct descended pairs. There are three cases:

- $\{a, b\} \cap \{c, d\} = \emptyset$: impossible by Lemma 9.
- $|\{a, b\} \cap \{c, d\}| = 1$ with distinct values: after relabeling, $\{a, b\}$ and $\{b, d\}$ with $a \ne d$, impossible by Lemma 10.

- $|\{a,b\} \cap \{c,d\}| = 1$ with a repeated element: $\{v,v\}$ and $\{v,w\}$, impossible by Lemma 11.

The only remaining possibility $\{a,b\} = \{c,d\}$ contradicts the assumption that the pairs are distinct. □

Corollary 12. *For any $k \geq 2$, every derived k-Lehmer factorization is produced by a unique plus-seed.*

6 Computational census for $k = 2$

Theorem 13 (Census)**.** *For $k = 2$, the following hold among all 2-Lehmer factorizations reachable through plus-seeds with at most six factors.*

(a) There are exactly 17 plus-seeds with $|F| \leq 6$ (including $\emptyset$). Six belong to the Fermat partial-product chain and eleven are sporadic (Table 1).

(b) The resulting 39 Lehmer factorizations include exactly 9 primitives (Table 2) and 30 derived factorizations.

(c) The number of derived factorizations produced by a plus-seed F equals the number of divisor pairs of $\Delta_+(N) = N^2 + N - 1$, where $N = \tilde{\epsilon}(F)$.

Proof. By exhaustive enumeration using the recurrence (3). Starting from $g(\emptyset) = 1$, factors are added with the constraint that g remains a positive integer at every non-terminal step. The search space is bounded by Hasanalizade's Theorem 1. The forest property is confirmed *a posteriori* (and now also *a priori* by Theorem 8). □

Table 1: Plus-seeds for $k = 2$ with $|F| \leq 6$.

$\lvert F\rvert$	Seed F	$\tilde{\epsilon}(F)$	Type
0	$\emptyset$	1	Fermat
1	(3)	3	Fermat
2	$(3,5)$	15	Fermat
3	$(3,5,17)$	255	Fermat
4	$(3,5,17,257)$	65 535	Fermat
	$(5,5,5,43)$	5 375	Sporadic
5	$(3,5,17,257,65537)$	4.29×10^9	Fermat
	$(3,5,17,353,929)$	8.36×10^7	Fermat-ext.
	$(3,9,9,23,131)$	7.32×10^5	Sporadic
	$(5,5,5,43,5377)$	2.89×10^7	Sporadic
	$(5,5,5,47,453)$	2.66×10^6	Sporadic
	$(5,7,7,7,133)$	2.28×10^5	Sporadic
6	$(3,9,9,23,131,732161)$	5.36×10^{11}	Sporadic
	$(3,11,11,11,677,3659)$	9.89×10^9	Sporadic
	$(5,5,5,53,215,158265)$	2.25×10^{11}	Sporadic
	$(5,7,7,7,133,228097)$	5.20×10^{10}	Sporadic
	$(5,7,7,13,13,619)$	2.56×10^7	Sporadic

Table 2: Primitive 2-Lehmer factorizations.

$\lvert G\rvert$	Primitive G	$\tilde{\epsilon}(G)$
5	$(5, 5, 5, 43, 5375)$	2.89×10^7
5	$(5, 5, 9, 9, 89)$	1.80×10^5
6	$(3, 5, 17, 377, 1217, 2295)$	2.69×10^{11}
6	$(3, 5, 17, 395, 1059, 2303)$	2.46×10^{11}
6	$(3, 5, 33, 53, 69, 8343)$	1.51×10^{10}
6	$(3, 9, 9, 23, 131, 732159)$	5.36×10^{11}
6	$(5, 5, 5, 47, 457, 50659)$	1.36×10^{11}
6	$(5, 5, 5, 65, 129, 2325)$	2.44×10^9
6	$(5, 7, 7, 7, 133, 228095)$	5.20×10^{10}

7 Plus-seed chains

The plus-seeds in our census do not arise in isolation: they organize themselves into infinite *chains* via a canonical extension operation. The Fermat partial products form one such chain, but they are not the only one. This section develops the chain theory and identifies its first sporadic instance.

7.1 The canonical chain extension

Theorem 14 (Chain extension exists and is unique)**.** *Let F^* be a $k = 2$ plus-seed of length s with $\tilde{\epsilon}(F^*) = E$ and $\tilde{\varphi}^*(F^*) = P$. Define $x := E+2$. Then (F^*, x) is again a $k = 2$ plus-seed. Moreover, x is the unique value of the next factor for which this is the case.*

Proof. For (F^*, x) to be a plus-seed we require $2P(x-1) = Ex+1$, i.e. $x(2P-E) = 2P+1$. Since F^* is itself a plus-seed, $2P-E = 1$ and $2P = E+1$, so the equation simplifies to $x = 2P+1 = E+2$. Substituting $x = E+2$ back, we have $2P(x-1) = (E+1)(E+1) = E^2+2E+1 = E(E+2)+1 = Ex+1$, so (F^*, x) is indeed a plus-seed. □

Lemma 15 (Chain extension is admissible)**.** *The next factor $x = E+2$ in Theorem 14 is odd, satisfies $x > x_s$ (the last factor of F^*), and respects the Lehmer congruence $x \not\equiv 1 \pmod{a}$ for every prior factor a in F^*.*

Proof. Parity. Each factor of F^* is odd (we restrict to the odd-spoof setting throughout), so E is odd and $E+2$ is odd.

Ordering. $E = \prod_i x_i \geq x_s$ since each $x_i \geq 1$, and in fact $E+2 > x_s$ for any non-trivial F^*.

Lehmer congruence. $x-1 = E+1 = 2P$. We need $a \nmid 2P$ for every prior factor a in F^*. Since a is odd, this is equivalent to $a \nmid P = \prod_j (x_j - 1)$. By the Lehmer congruence on F^* itself, $a \nmid (x_j - 1)$ for all j with x_j a factor of F^* (taking j to be the index of a for the trivial case, where $a \nmid (a-1)$ since $a \geq 3$). Hence $a \nmid P$. □

Corollary 16 (Plus-seed chains are infinite)**.** *Every $k = 2$ plus-seed F^* generates an infinite chain*

$$F^* \;\subset\; F^* \cdot (E_0 + 2) \;\subset\; F^* \cdot (E_0 + 2,\, E_1 + 2) \;\subset\; \cdots$$

of $k = 2$ plus-seeds, where E_i is the value of $\tilde{\epsilon}$ at the i-th step.

7.2 The Fermat chain

The Fermat partial products $F^{(s)} = \prod_{i=0}^{s-1}(2^{2^i}+1)$ form a linear chain of plus-seeds for $k = 2$, since the telescoping identity

$$2\prod_{i=0}^{s-1} 2^{2^i} = 2^{2^s} = \prod_{i=0}^{s-1}(2^{2^i}+1)+1$$

gives $2\,\tilde{\varphi}^*(F^{(s)}) = \tilde{\epsilon}(F^{(s)})+1$. Each seed generates Lehmer factorizations by factoring $\Delta_+(2^{2^s}-1) = 2^{2^{s+1}} - 2^{2^s} - 1$.

This linear chain is precisely the chain that Theorem 14 produces from the root (3): $E^{(s)} = F_s - 2 = 2^{2^s} - 1$ (by telescoping), so $E^{(s)} + 2 = 2^{2^s} + 1 = F_s$, and the canonical chain extension yields the next Fermat number.

Corollary 17 (Fermat chain identification)**.** *The chain extending* (3) *via Theorem 14 is the sequence of Fermat partial products* $F^{(s)} = (F_0, F_1, \ldots, F_{s-1})$ *for* $s \geq 1$.

7.3 The first sporadic chain

The Fermat chain is not the only one. Our $|F| \leq 6$ census exhibits a sporadic plus-seed at length 4:

$$S^{(4)} = (5, 5, 5, 43).$$

By Theorem 14 this extends to an infinite chain

$$S^{(4)},\ S^{(5)} = (5, 5, 5, 43, 5377),\ S^{(6)} = (5, 5, 5, 43, 5377, 28901377),\ \ldots$$

where $5375 + 2 = 5377$ and $5375 \cdot 5377 + 2 = 28901377$, etc.

The Fermat-prefix closedness theorem of the companion paper [3] extends to this and to every plus-seed:

Lemma 18 (Coprimality lemma for plus-seeds)**.** *For any* $k = 2$ *plus-seed* $F^* = (x_1, \ldots, x_s)$ *with* $E = \tilde{\epsilon}(F^*)$ *and* $M_{F^*} := E^2 + E - 1$*, we have* $\gcd(M_{F^*}, x_i) = 1$ *for every* $i \in \{1, \ldots, s\}$.

Proof. $E = \prod_j x_j$ contains x_i as a factor, so $E \equiv 0 \pmod{x_i}$. Hence $M_{F^*} = E^2 + E - 1 \equiv -1 \pmod{x_i}$, and $\gcd(M_{F^*}, x_i) \mid \gcd(-1, x_i) = 1$. □

Theorem 19 (Plus-seed closedness at $r = s + 2$)**.** *Let* F^* *be a* $k = 2$ *plus-seed of length* s*, with* $E = \tilde{\epsilon}(F^*)$ *and* $M_{F^*} = E^2 + E - 1$*. The* $k = 2$*-Lehmer factorizations of length* $s + 2$ *extending* F^* *are in bijection with the divisors* $d \leq \sqrt{M_{F^*}}$ *of* M_{F^*}*. The factorization corresponding to* d *is*

$$\big(F^*,\ d + (E+1),\ M_{F^*}/d + (E+1)\big),$$

and the count is exactly $\tau(M_{F^*})/2$.

Proof. By the $s = r - 2$ finishing-feasibility identity (Proposition 2 of [3]) applied to F^*, the length-$(s+2)$ extensions are in bijection with divisor pairs $(d, M_{F^*}/d)$ of $M_{F^*} = B \cdot E - A = (E+1) \cdot E - 1$, with $a = d + B$ and $b = M_{F^*}/d + B$ where $B = 2P = E + 1$ for any plus-seed. Integrality follows from $A = 1$. The Lehmer congruence (i.e. $x_i \nmid (a-1)$ and $x_i \nmid (b-1)$ for prior factors x_i) holds because $a - 1 = d + E$ and $E \equiv 0 \pmod{x_i}$ implies $a - 1 \equiv d \pmod{x_i}$, and by Lemma 18, $x_i \nmid d$ (since $d \mid M_{F^*}$ and $\gcd(M_{F^*}, x_i) = 1$). The same applies to $b - 1$. Parity holds because E is a product of odd factors, hence odd, so $E^2 + E$ is even and $M_{F^*} = E^2 + E - 1$ is odd; all its divisors are odd. The remaining ordering and $a \mid b - 1$ checks are as in the Fermat case [3, Theorem 3]. □

The theorem is verified empirically by the $r \leq 7$ census: for each of the eight $k = 2$ plus-seeds with length ≤ 5, the predicted count $\tau(M_{F^*})/2$ matches the number of length-$(s+2)$ extensions found by exhaustive search:

$(3, 5, 17, 257)$	$\tau/2 = 4$	✓
$(5, 5, 5, 43)$	$\tau/2 = 8$	✓
$(3, 5, 17, 257, 65537)$	$\tau/2 = 16$	✓
$(3, 5, 17, 353, 929)$	$\tau/2 = 4$	✓
$(3, 9, 9, 23, 131)$	$\tau/2 = 1$	✓
$(5, 5, 5, 43, 5377)$	$\tau/2 = 1$	✓
$(5, 5, 5, 47, 453)$	$\tau/2 = 8$	✓
$(5, 7, 7, 7, 133)$	$\tau/2 = 4$	✓

7.4 Fresh plus-seeds and the chain forest

We say a plus-seed of length $s \geq 2$ is a *descendant* if its length-$(s-1)$ prefix is itself a plus-seed; otherwise it is *fresh*. By Theorem 14, each fresh plus-seed initiates a unique infinite chain of descendant plus-seeds. The set of all $k = 2$ plus-seeds, organized under the parent-child relation $F^* \mapsto (F^*, E+2)$, forms a disjoint union of infinite linear chains rooted at the fresh plus-seeds.

Table 3: Plus-seed structure by length, $k = 2$, odd spoof. Each fresh plus-seed initiates an infinite descendant chain by Corollary 16; descendant plus-seeds are not listed.

length s	total plus-seeds	fresh	fresh plus-seeds at this length
1	1	1	(3)
2	1	0	—
3	1	0	—
4	2	1	$(5, 5, 5, 43)$
5	6	4	$(3, 5, 17, 353, 929)$, $(3, 9, 9, 23, 131)$, $(5, 5, 5, 47, 453)$, $(5, 7, 7, 7, 133)$

Remark 20 (Two children per plus-seed). A length-s plus-seed F^* produces two distinct length-$(s+1)$ descendants:

- the *Lehmer descendant* (F^*, E), satisfying $2P(E-1) = E \cdot E - 1$, which is a $k = 2$-Lehmer factorization by Proposition 2 of [3] applied at the plus-seed; and
- the *plus-seed descendant* $(F^*, E+2)$ from Theorem 14.

The Lehmer descendant is the bottom of an extension tree (no further extension); the plus-seed descendant continues the chain. This dual structure—"one length-$(s+1)$ Lehmer companion plus one length-$(s+1)$ plus-seed continuation"—governs the local structure of $\mathcal{G}_2$ at every plus-seed.

8 Open questions

1. **Finiteness of primitives.** All nine known primitive 2-Lehmer factorizations have $|G| \geq 5$. Are there finitely many for each k?

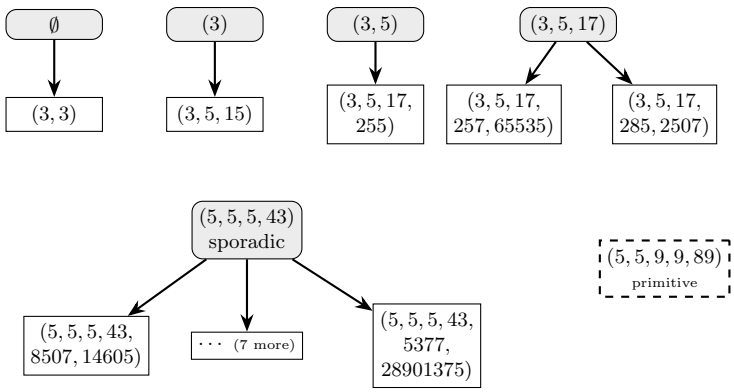

Figure 1: A fragment of $\mathcal{G}_2$. Rounded: plus-seeds. Solid: derived Lehmer factorizations. Dashed: primitive. The $(5, 5, 5, 43)$ subtree exhibits both children: the Lehmer descendant $(5, 5, 5, 43, 5375)$ via $x = E$, and the chain extension $(5, 5, 5, 43, 5377)$ via $x = E + 2$.

2. **Realizability.** Which spoof factorizations correspond to actual integers $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with each $x_i = p_i^{\alpha_i}$ a prime power?

3. **Sporadic seed structure.** Is there a generating mechanism for the sporadic plus-seeds, or a secondary recursive structure explaining their proliferation?

4. **Density of primitives.** Among our 39 factorizations, 9 are primitive (≈23%). Does this proportion converge?

References

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